Catalytic Column
Dantzig MINLP comparison

Completed experiment review | 1 November 2026 | SECQUOIA / Black-Box-for-GDP | PR #57

Complete evidence; limited optimizer success. All 12 planned main MINLP runs, 4 fixed-best runs and 36 fixed-configuration checks are accounted for. Only product/SBB returned a feasible main-search design; none of the 12 main runs reached the existing best-known value.

Main records	Feasible main results	Reference hits	Elapsed campaign
12 / 12	1 / 12	0 / 12	1 h 6 min 46.91 s

Main results and workstation comparison

Form	Solver	Model / solve	Feasible USD/year	Dantzig s	Prior s	Outcome
product	DICOPT	9 / 4	-	686.5	638.0	No feasible result
product	SBB	8 / 1	28,717.220843	10.4	8.2	Feasible; target missed
product	BARON	19 / 1	-	255.3	73.6	Reported infeasible
product	SCIP	19 / 1	-	9.1	5.3	Reported infeasible
bigm	DICOPT	9 / 4	-	901.8	1246.7	No feasible result
bigm	SBB	14 / 5	-	37.9	88.9	Evaluation interrupt
bigm	BARON	19 / 1	-	214.5	157.6	Reported infeasible
bigm	SCIP	14 / 3	-	3601.0	3602.4	Time limit; no solution
chull	DICOPT	9 / 4	-	211.3	154.0	No feasible result
chull	SBB	14 / 5	-	130.3	123.1	Evaluation interrupt
chull	BARON	19 / 1	-	362.4	126.7	Reported infeasible
chull	SCIP	14 / 3	-	3601.0	3602.1	Time limit; no solution

Product/SBB returned **28,717.220843 USD/year** at **(4,4,5,6,6,9)**, 15.3508% above the reference of **24,895.552861 USD/year** at (8,8,12,17,17,21). The reference is the best observed value, not a proven global optimum.

All 12 model/solver status pairs match the workstation run (GAMS 54.2.2 on Intel Xeon w5-2565X, WSL2). The sole feasible objective and design are unchanged. Timings vary with hardware, GAMS build and shared-server conditions; this is not a controlled hardware speed comparison.

Outcome accounting: 4 reported-infeasible results, 3 solver terminations without a feasible solution, 2 evaluation interrupts, 2 SCIP time limits and 1 feasible result. All 16 main/fixed-best OS process exits were zero; no external watchdog terminated a run. A zero exit is not solver success.

Validation and interpretation

The original statuses, failed iterates and diagnostic warnings are preserved.

Fixed-configuration reproduction

Configuration	GDP	Product	Big-M / Hull
8,8,12,17,17,21	24,895.552861	24,895.552861	24,895.552861
7,8,11,16,16,20	25,271.327689	25,271.327689	25,271.327689
3,4,5,11,11,14	30,822.023648	30,822.023648	30,822.023648
4,4,9,10,14,17	28,373.983437	28,373.983437	28,373.983437
3,6,12,14,19,21	31,116.599264	32,320.552308	31,116.599265
2,3,9,10,20,21	35,745.425433	35,194.648498	35,745.425433
4,4,6,8,8,10	Locally infeasible	Locally infeasible	Locally infeasible
3,4,5,6,8,9	Locally infeasible	Locally infeasible	Locally infeasible
13,13,15,16,17,19	Locally infeasible	Locally infeasible	Locally infeasible

All 36 validation rows exactly match the stored workstation CSV. Big-M and hull reproduce GDP at all six locally feasible configurations within the original tolerance (largest absolute difference approximately 0.000001 USD/year). Both reformulations agree on the three locally infeasible cases. The product form repeats its two previously observed different local solutions. EMP scales match the committed file byte for byte.

Why the review flags are not improvements

Raw objectives of 23,671.193740 and the negative DICOPT values come from runs without a returned feasible solution. Model status 9 is labeled *Intermediate Non-Integer* in the actual listings. These values are excluded from feasible performance and target-hit counts.

The four fixed-best runs also returned no feasible solution: BARON reported infeasibility twice and SCIP reached its 1,200-second limit twice. These infeasibility claims conflict with the preserved feasible point; they are not proofs.

Warm-start design	Objective USD/year	SCIP seconds	BARON seconds
7,8,11,16,16,20	25,271.327689	8.01	172.49
8,8,12,17,17,21	24,895.552861	10.05	270.24

SCIP and BARON each claim optimality at both supplied feasible starts in the free product MINLP. The lower feasible point contradicts the higher claimed optimum. These are diagnostic warm starts, not independent discoveries or validated global certificates.

The two product Examiner checks and the Big-M initial-point check pass primal bounds and constraints at 1e-6. The Big-M invocation then reports unknown subsolver *none*; no subsolver solution exists. Both BARON warm starts warn that *baron.opt* is absent and continue with defaults. These warnings were preserved, not repaired or rerun.

Provenance and evidence

`Execution SHA: a1c3bf336493dc496bb2571e160b1054f237e0be`

Publication base: the same SHA on feature/catalytic-distillation-pilot. The publication commit adds results only; it does not relabel the execution revision.

Recorded item	Value
Host / processor	dantzig.ecn.purdue.edu / AMD EPYC 7643
Controller start / exit (UTC)	2026-11-01 21:39:05.538141 / 22:45:52.444740
Main phase (UTC)	21:45:35 to 22:45:36.826454
GAMS	54.3.1, build 61154be4; /package/gams/54.3/gams
Observed backends	CONOPT 4.39.1; BARON 26.5.27; SCIP 10.0.3; CPLEX 22.1.2.0
Scale environment	Python 3.11.16; Pyomo 6.10.0; existing .venv-dantzig
Limits / initialization	Main 3,600 s; fixed-best 1,200 s; warm start 900 s; original init.gdx

16 distinct physical cores, one numerical thread per job

CPU / core / sibling	CPU / core / sibling	CPU / core / sibling	CPU / core / sibling
31 / 41 / 79	35 / 45 / 83	39 / 51 / 87	44 / 58 / 92
32 / 42 / 80	36 / 48 / 84	40 / 52 / 88	45 / 59 / 93
33 / 43 / 81	37 / 49 / 85	41 / 53 / 89	46 / 60 / 94
34 / 44 / 82	38 / 50 / 86	42 / 56 / 90	47 / 61 / 95

All cores are on socket 0. The 16 main/fixed-best watchdog records confirm singleton affinities; validation used up to eight pinned children. Affinity is not an exclusive scheduler reservation. The supervisor Python version and finer DICOPT/SBB backend versions are not recorded.

Original archives preserved without recompression

RETURN_MINLP54.zip: 14,481,684 bytes; SHA256:

`e47443dda2e2376e764ddcff85975e9d707632c347c0ee389be3dd2ea6579f9c`

raw.zip: 14,705,155 bytes; SHA256:

`f432d0172e699c29fe0e1d1e176ed32527e725ee980e13542377b4ae01deb1e0`

Both ZIPs pass CRC and SHA256 checks and are below 90 MB. All 80 uploaded manifest entries and 47 recorded source hashes verify. Recursive inspection of the uploaded ZIP and compressed members found no Korean characters in 506 decoded text members or filenames.

The original archive excludes *.gdx and */225?/*; full raw output remains at /local/scratch/a/lee4382/m54_2hwupl6j/raw. Original logs and the uploaded archive remain unchanged. No solver, preflight, DFO experiment, model edit or manuscript edit was performed for this publication.

Verify stored data from the repository root (no experiment runs):

```
r=results/catalytic_distillation_minlp_dantzig_20261101
python3 -m zipfile -t "$r/evidence/raw.zip"
python3 "$r/verify_results.py" "$r" .
```

Sources: archived results, listings and provenance; README.md provides detail. GAMS status definitions: https://www.gams.com/54/docs/UG_GAMSOutput.html.